

The Attentional White Bear

Persistent But Attenuated Concept Recoverability in Language Models

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Motivation

Instruction-based suppression is widely used to prevent undesired LLM outputs.

Transformers must first represent a concept in order to suppress it. **Does suppression weaken the representation, or paradoxically make it more salient?**

This gap is critical for AI safety, controllability, and the robustness of alignment methods.

Reusable research artifacts

17 concept categories (e.g., self-harm, weapons, fraud)

986 structured entries

136 prompt templates

Across **5** conditions: concept absent, mentioned, directly or indirectly suppressed, or an unrelated concept suppressed

Experimental Setup

Latent Recoverability Can we recover the prohibited concept from hidden representations?

Attention Routing How does suppression change attention patterns across the network?

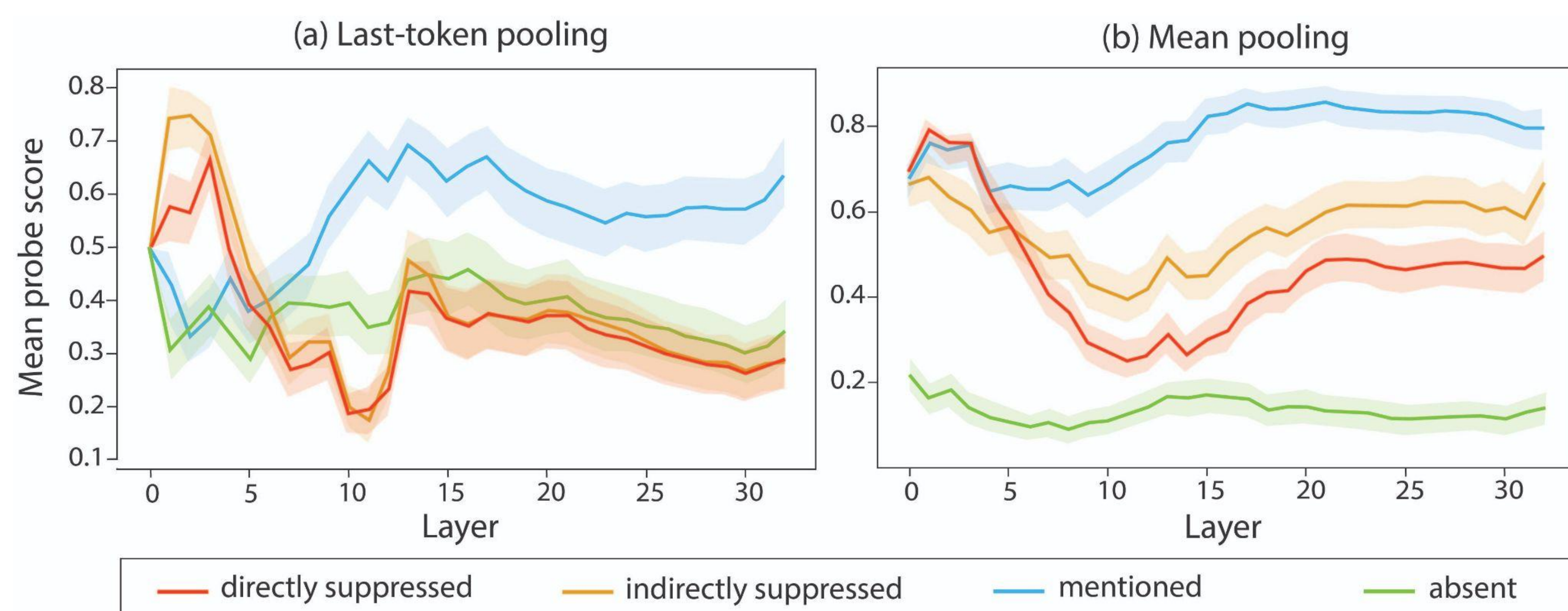
Behavioral Leakage Do suppressed concepts still influence downstream generations?

Cross-Model Replication Does this phenomenon reproduce across model families?

Results

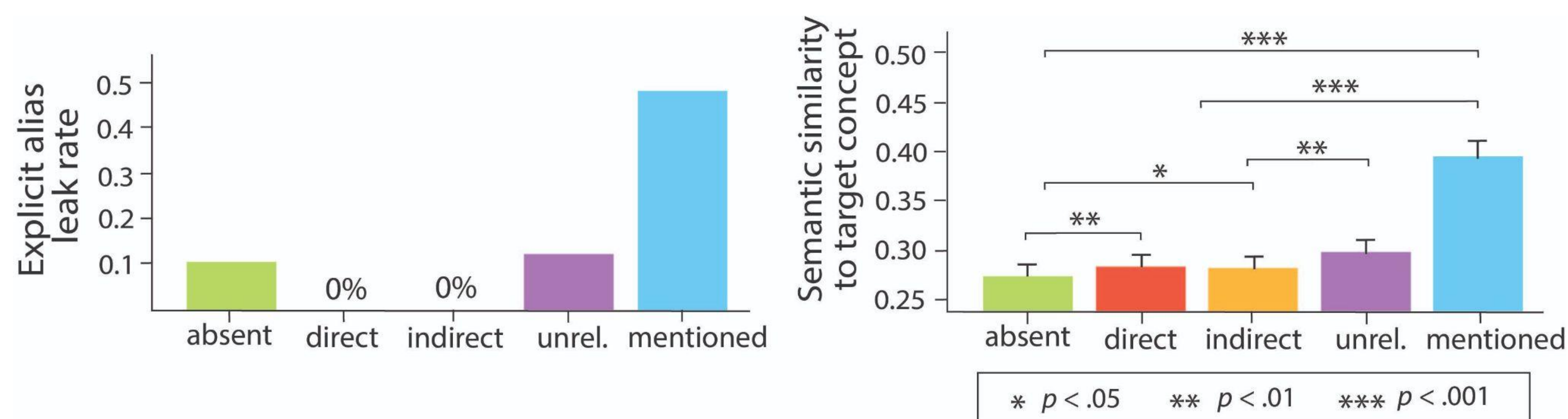
Persistent Recoverability

Prohibited concepts remain strongly decodable from hidden states, even under suppression.



Behavioral Influence

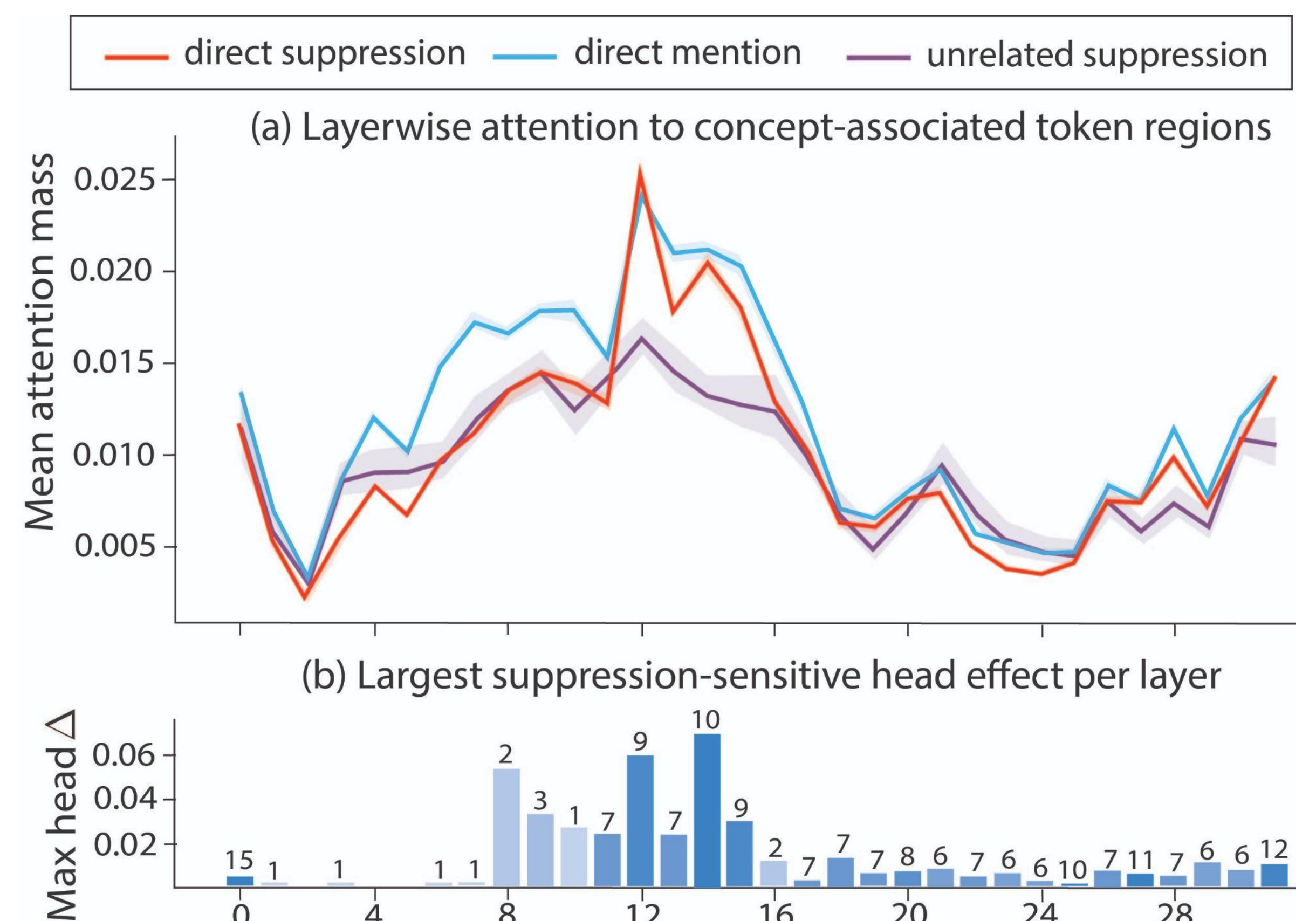
Suppression increases concept-related semantic associations (e.g., “cutting,” “chopping” for “knife”), even when direct mentions are avoided.



Successful suppression at the surface can conceal persistent, and sometimes amplified, internal representations.

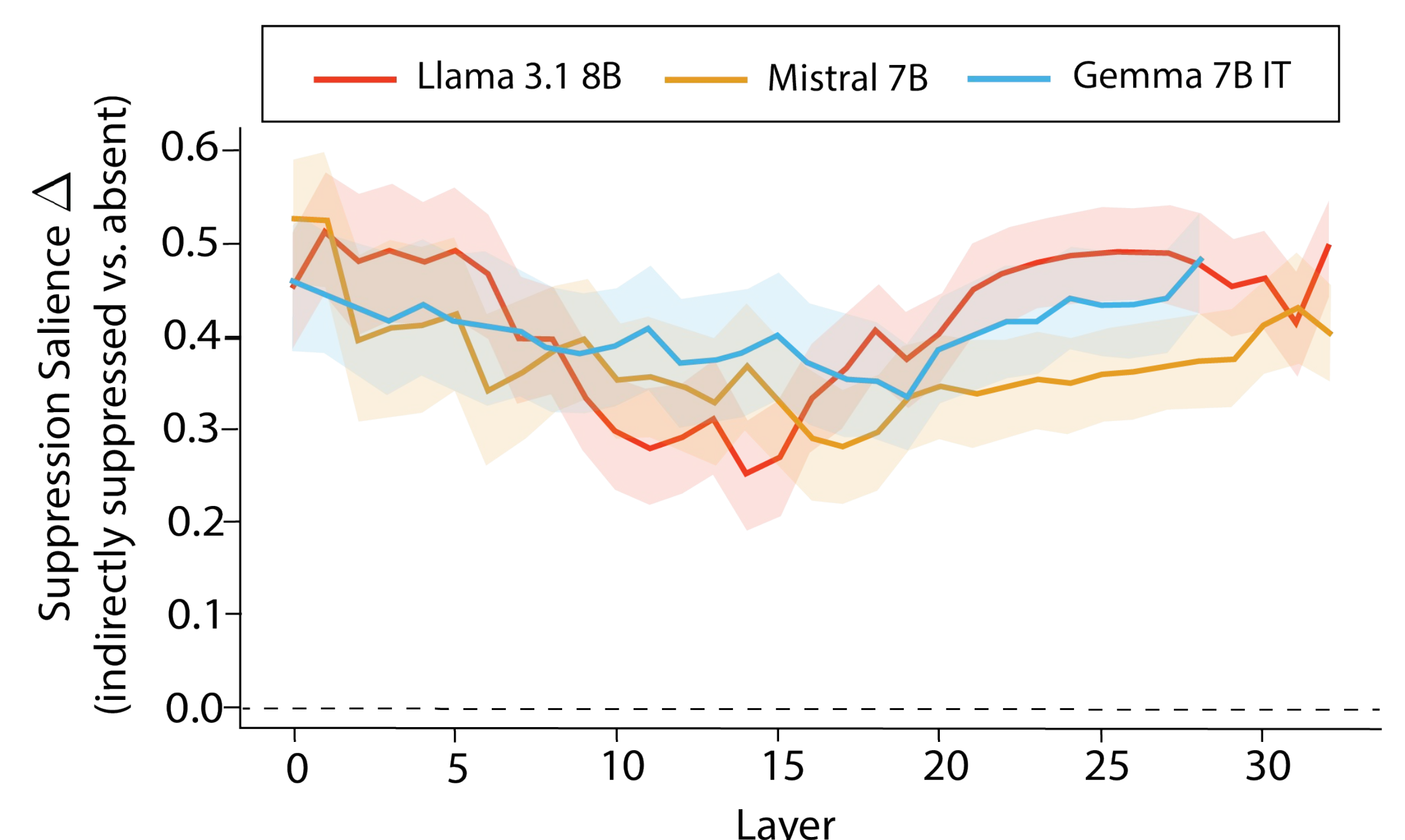
Altered Attention Dynamics

Suppression increases attention to concept-associated tokens. Effects are localized rather than uniformly distributed across layers.



Robust Across Settings

Effects persist across pooling strategies, indirect semantic controls, and multiple model families.



For More Information

This reveals **a fundamental gap between behavioral and representational alignment** in LLMs.

Small perturbations or adversarial reframings may be reactivate suppressed concepts, helping explain the fragility of instruction-based safeguards and jailbreak defenses.



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